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BANK CUSTOMER CLUSTERING USING A WARD METHOD: EMPIRICAL ANALYSIS OF TBILISI, GEORGIA

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Abstract: *The purpose of this paper is to study the bank consumer behavior and characteristics, to divide them into homogeneous clusters and make a group of several data that has characteristics as closely as possible in order to help commercial banks to better identify their customers and design more efficient marketing strategies. Data from 220 bank customers were analyzed. The Ward method has been used to group the cluster, using the Euclidean distance squared and performing standardization by variable in Z scores to avoid distortions between different units of measurement. In the author's knowledge, there are not available similar data that focus on individuals' behavior and characteristics on the mortgage loan market in Georgia. The author believes that this paper will gain some attention from commercial banks and policymakers and can allow them to make the market safer and competitive. The cluster analysis indicates that married individuals have more income and co-pay the loan and women are associated with greater risk aversion. The type and location of housing are determined by having children. The study can be used by regulators, financial and public institutions like commercial banks to increase market security, offer different products to the customers, and focus their efforts on the best and most profitable consumers. The author believes that the study has the merit of offering different ideas for further research.*

Keywords*: housing, consumer economics, mortgages, cluster analysis*

INTRODUCTION

Residential property serves as a basic human need and plays a significant part in the urban economy (Suhaida, Tawil, Hamzah, Che Ani, & Tahir, 2010). International human rights law recognizes everyone's right to an adequate standard of living, including adequate housing, which was recognized as part of the right to an adequate standard of living in the 1948 Universal Declaration of Human Rights and in the 1966 International Covenant on Economic, Social and Cultural Rights (UN-habitat, 2014).

Georgia is known for its architecture of traditional dwellings, which shows that it is very important for the Georgian people to own their housing (Meparishvili, 2016). Unfortunately, there is still no database that would characterize homeowners in Georgia in terms of gender, age, marital status, etc. In Georgia, people are very rooted in housing. In general, children live with parents, mostly before they finish university, and after they find a stable job in the public or private sector, they decide to take a mortgage and buy housing and think about the marriage. The youth is flocking to cities in record numbers in pursuit of career opportunities and a better lifestyle. As a result of this urbanization and Westernization, demand for apartments has increased (Brown, 2020). Georgians prefer to own, rather than rent. The cultural characteristic that owning a flat is the best achievement and the best investment decision (Bochorishvili & Samkurashvili, 2019).

In 2018, Georgian Geostat conducted the Pilot Survey on Measuring Asset Ownership and Entrepreneurship from a Gender Perspective (Tsekvava, Kalakashvili, Gulua, Paksashvili, & Tvalodze, 2018). The main findings of the pilot survey are that dwelling has the highest incidence of ownership among core assets.

Housing properties are in stronger demand as population increases (Shuid, 2016). Credit debt of households in Georgia continues to grow at a rapid pace, which is noteworthy in terms of overindebtedness. Greater financial debt would affect the health of the entire economy (Kroszner & Strahan, 2014). Consumer loans, mortgage loans, business loans and auto loans are among the major types of consumer credit in Georgia. Therefore, the growing demand for mortgage loans requires systematic research.

This paper analyses published literature on the similar market with the usage of cluster analysis, describes the research methodology, discusses the main results and ends by summarizing the concluding remarks.

LITERATURE REVIEW

Researchers argue that customer segmentation is a crucial practice of the customers' relationship management, which enables businesses to achieve a deeper understanding of customers' needs and behaviors (Abbasimehr & Shabani, 2019). Mortgage choices made at loan origination may help reveal information about homeowners' decision staying in the house and their stage in the lifecycle. Mortgage information relates to seller characteristics, especially the individuals expected tenure in the house (Pace & Zhu, 2018).

Since data stored in the databases of banks are generally complex and are aimed at varying dimensions such as consumer loan, debt collection, credit reporting and mortgage, the procedure for data analysis becomes complicated.

Bank databases store huge volume of information such as the issues given by the customers. The ultimate goal of any business is to increase the number of customers and the quality of services rendered to them and get their loyalty (Sundar, Narmadha, Jebapriya, & Malathy, 2019).

Customer segmentation has an important role in customers' relationship management, because it allows companies to design and establish different strategies to maximize the value of customers. Customer data analysts can promote appropriate marketing strategies that are more profitable (Hung, Lien, & Ngoc, 2019).

Demands for different types of loans may differ among families with various structures, which can be categorized in terms of marital status, gender of household head, and presence of children (Xiao & Yao, 2019). Demographic data, monthly income of households and individual account characteristics are also used in some papers, where researchers collect information from credit bureaus.

RESEARCH METHODOLOGY

The data used for the analysis are from an own survey on mortgage loan consumer which was conducted in Tbilisi (Georgia) in July-August, 2019. Based on the review of empirical and theoretical literature, a quantitative questionnaire was developed including different variables (age, household size, marital status, income, job, sex). Structured interview method was chosen for the study, since the content of the questionnaire is the same for each respondent, which allows for aggregation of responses (Bryman & Bell, 2011).

The population was the number of mortgage borrowers. According to the National Bank of Georgia, the number of mortgage loans granted to households in Tbilisi in August 2019 was 61,000. This population has to meet the sampling size of 382 respondents to generate accurate results (Krejcie & Morgan, 1970). In this research, there are 220 questionnaire forms that are returned and can be used for cluster analysis.

Four interviewers conducted face-to-face interviews mainly in front of the bank branches during the working hours to maximize the exposure of mortgage loan customers. According to the National Bank of Georgia, the number of licensed banking institutions in Georgia was 15 units in 2019. Nevertheless, the survey was conducted mainly in front of the two largest banks – in the branches of the TBC Bank and the Bank of Georgia.

A simple random sampling method was chosen, since at that time each element of the generic set has a known and equal chance of being selectively aggregated (Malhotra, Nunan, & Birks, 2017). Interviewers asked every third

customer whether they had a mortgage loan or not. The similar data are not still available for Georgian mortgage market. In contrast, most of the data are stored by commercial banks for their internal and private use. *Table 1* shows the technical data of the survey.

Table 1

Technical data of the study

Universe	*Individuals aged 18-65 years*
Information collection	Face-to-face interviews using a structured questionnaire
Sample selection	Simple sampling method. Consumers of the mortgage loans are considered as the sampling unit
Sample size	220 individual face-to-face interviews
Sampling error	The sampling error for the sample as a whole is ±5% for a confidence level of 95% and assuming p = q = 0.5
Fieldwork dates	From July 2019 until August 2019

Source: developed by author

Ward and Joe (1963) proposed that the loss of information that occurs when integrating different individuals into clusters could be measured by the sum total of the squares of the deviations between each point (individual) and the mean of the cluster in which it is integrated. In order for the clustering process to be optimal, in the sense that the groups formed do not distort the original data, it is proposed the following strategy: *in each step of the analysis, consider the possibility of joining each pair of groups and opt for the merger of those two groups that least increase the sum of the squares of the deviations when joining.*

Ward's method is one of the most widely used in practice; it has almost all the advantages of the mean method and tends to be more discriminating in determining clustering levels. Research by Kuiper and Fisher (1975) proved that this method was able to match the optimum classification better than other methods (minimum, maximum, mean and centroid).

RESULTS

Based on mortgage loan customers we can characterize each cluster.

In clusters 1 and 2, we can observe young consumers, however in their decision on the location the level of income and the quality of the same intervenes directly. Consumers in cluster 1, who belong to the private sector, have greater economic security and potential. The higher the income, the better the location, i.e. central district. The same applies to cluster 5, except that they are single. This idea coincides with the findings of Bradbury and Chalmers (2003). Clusters 3 and 4 are composed of women. Cluster 3 is focused on young women without children and with average incomes, who choose to live away from the central district. This is probably because they

cannot afford another location. However, cluster 4 is composed of mature married women with children who, despite having a high income, prefer locations away from the central district. It is because of the lifestyle offered by certain suburban neighborhoods and because they can afford larger houses with gardens and playgrounds for the children. This same trend occurs with cluster 6, we find the same age group, with high incomes and children, again choosing the location in the away from central district areas.

In view of the results, we could assume that the opinion of the woman is relevant in the decision of the location; likewise, the factor to have children is decisive to live away from the central district. These results could be explained by the greater risk aversion associated with women (Jurajda & Janhuba, 2016). While the marital status is not influential in that sense.

Below are the clusters of Bank of Georgia customers.

Clusters 1 and 5 are composed of foreign citizens, who are married and with high incomes. They own private businesses and therefore can afford a luxury home with a mortgage loan in the central district of the city. This is because doing business in Georgia is very easy. According to the doing business database of 2019, Georgia is among the top ten countries in the world (The World Bank, 2020).

In clusters 2, 3, and 6 we can observe also private business owners, but they are Georgian citizens. Because of the higher income and family size, they prefer central districts.

Cluster 4 consists of women who are married and have children. Half of them work in the public sector and others in the private sector. They prefer to buy housing outside the city center. Because children's entertainment venues are mostly located at the exit of the city. In addition, there is more recreational space away from the central district.

CONCLUSIONS

In Georgia, there are no prior banking studies based on cluster analysis, because the data on customers is stored in commercial banks and it is impossible to do an empirical analysis. Therefore, this is one of the first attempts. In this study, we present the two largest commercial bank customer clustering using a Ward method.

Our results show that TBC bank customers are more or less young compared to the Bank of Georgia. The different strategic choices of TBC Bank should be more tailored to the younger generations. In modern Georgia, the average family size is gradually declining as young people seek for individual housing. Accordingly, both banks should be focused on young individuals and offer them flexible terms such as student loans, preferential period mortgage loans, and so on.

Our study shows that married individuals have more income and co-pay the loan. Therefore, the clustering family structure gives helpful information for commercial banks, in terms of identifying risk factors. Consequently, when commercial banks issue loans to consumers, they should take into account the perception that the behavior of married and unmarried individuals is different. That is why they have to offer a variety of marketing products or services like a student loan, car insurance, house insurance, retirement pension. The bank's future depends on the number of different products/services they sell together with the mortgage loan. In addition, commercial banks should offer extra insurance for mortgage loan holders to protect them from the unscrupulous behavior of construction companies such as fraud, unreliability and construction incompleteness.

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